Understanding the Dataset

Credit Card Approval Prediction System | Epic 3: Dataset Download and Understanding

Objective

Explore data quality, distributions, and the target variable derivation logic before selecting features for the classification model.

Code: Exploration

```
application_df.info()
application_df.describe()
application_df.isnull().sum()

credit_df['STATUS'].value_counts()
```

Missing Values — application_record.csv

Column	Missing	% Missing	Action
OCCUPATION_TYPE	134,203	30.6%	Fill with 'Unknown'; pipeline uses handle_unknown='ignore'
All other columns	0	0%	No action needed

Credit STATUS Distribution

STATUS Code	Meaning	Count	Label
C	Paid off / closed	442,031	TARGET = 0 (Low Risk)
0	1–29 days late	383,120	TARGET = 0 (Low Risk)
X	No loan this month	209,230	TARGET = 0 (Low Risk)
1	30–59 days late	11,090	TARGET = 1 (High Risk)
5	150+ days overdue	1,693	TARGET = 1 (High Risk)
2	60–89 days late	868	TARGET = 1 (High Risk)
3	90–119 days late	320	TARGET = 1 (High Risk)
4	120–149 days late	223	TARGET = 1 (High Risk)

Target Variable Derivation

```
def create_target(statuses):
    if any(s in ['1','2','3','4','5'] for s in statuses):
        return 1  # Rejected — High Risk
    return 0      # Approved — Low Risk

target_df = (
```

```
    credit_df
    .groupby('ID')['STATUS']
    .apply(list)
    .reset_index()
)
target_df['TARGET'] = target_df['STATUS'].apply(create_target)
```

Model Comparison Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	~0.72	~0.68	~0.70	~0.69	~0.79
Decision Tree	~0.82	~0.80	~0.82	~0.81	~0.82
Random Forest	~0.89	~0.88	~0.87	~0.87	~0.95
XGBoost	~0.88	~0.87	~0.86	~0.86	~0.94

Random Forest was selected as the best model based on highest Accuracy, F1-Score, and ROC-AUC across all candidates.